Lattice Resonance in a Connected Algebraic Action: Integer Covolumes and a Finite-Index Correction

HCS-C388 source-theorem and reproducibility package

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Abstract

We compute the exact component correction for the resonant fixed groups of the connected algebraic action defined by $1 + u + v$. A general integer-image covolume lemma converts the product of nonzero singular values into the torsion order of a cokernel. The saturated two-mode kernel has Gram determinant $N^2/3$ at lattice index N, so the nonzero Fourier product must be multiplied by $3/N^2$. The index-three all-ones matrix gives an exact minimal counterexample to the uncorrected finite-lattice formula printed in an explicitly identified arXiv version. This statement does not refute the classical entropy asymptotic or establish the status of later corrigenda. Exact Smith witnesses and characteristic coefficients provide reproducible finite controls for the all-lattice proof.

中文摘要

本文给出上述连通代数作用的共振固定群分支数修正。 一般整数像格的体积引理把非零奇异值乘积转化为余核挠子群的阶。
在指数为正整数的子格上，饱和整数核格的格拉姆行列式精确等于指数平方的三分之一，因此必须加入相应倒数修正。
最小指数三的全一矩阵给出所核对预印本版本中有限格公式的精确反例。
这个局部修正不否定经典熵渐近定理，也不声称已确定后续勘误情况。
完整史密斯证书和特征多项式为全子格证明提供独立有限核验。

Keywords: algebraic action; lattice resonance; Smith form; integer covolume; Mahler measure; orbit counting
关键词：代数作用；子格共振；史密斯形；整数格体积；马勒测度；轨道计数

1 One relation, two different kinds of fixed groups

A Fourier zero is not merely an eigenvalue to omit from a determinant. For a compact connected algebraic action, it can create a continuous family of periodic states. The relation studied here is elementary:

$$X = \{x \in \mathbb{T}^{\mathbb{Z}^2} : x_{i,j} + x_{i+1,j} + x_{i,j+1} = 0\}, \qquad (\alpha^{(a,b)}x)_{i,j} = x_{i+a,j+b}, \quad \mathbb{T} = \mathbb{R}/\mathbb{Z}. \tag{1}$$

The challenge addressed by this reconstruction is to retain the integral lattice while passing between fixed groups, Fourier modes and finite component counts. Real diagonalization alone discards that information.

Every acting sublattice has a unique column Hermite normal form

$$\Lambda = \langle (a, 0), (b, c) \rangle, \qquad a, c > 0, \quad 0 \le b < a, \quad N = [\mathbb{Z}^2 : \Lambda] = ac. \tag{2}$$

We write Fix_Λ for points fixed by every element of Λ, not for points whose stabilizer is exactly Λ. The central source dichotomy is finite versus a finite union of two-tori. The additional finite-lattice statement is that the resonant component count requires the factor $3/N^2$. For example, at the rectangular 3×3 quotient the nonzero Fourier product is 81 but there are only three components. At the smallest skew index-three quotient the product is 3 and there is one component.

Classical ownership and the precise scope. Lind, Schmidt and Ward identify the entropy of principal algebraic actions with Mahler measure, and display this exact polynomial in their introduction [1]. Smyth owns its special-value evaluation [2]. Lind, Schmidt and Verbitskiy give the exact Hermite congruence and the finite-versus-two-torus distinction in Example 3.2 of the accessed source [3]. None of those facts is presented here as a literature discovery. The package supplies an explicit all-lattice reconstruction, complete integer certificates and a careful distinction between the quantities that an orbit-count construction needs. Section 5 identifies a finite-lattice error in the displayed formula of one accessed version. It neither infers the status of later corrigenda nor challenges the entropy asymptotic from that error.

2 Connected source and exact quotient matrices

The relation in (1) is closed in a product of circles, so X is a compact metrizable abelian group. Coordinate shifts are automorphisms. Its discrete character module is

$$\widehat{X} = \mathbb{Z}[u^{\pm 1}, v^{\pm 1}]/(1 + u + v) \simeq \mathbb{Z}[u^{\pm 1}, (1 + u)^{-1}]. \tag{3}$$

The right-hand ring embeds in $\mathbb{Q}(u)$ and is additively torsion-free. The duality criterion for compact abelian groups therefore makes X connected. This is not a finite-characteristic three-dot shift: its coordinates are circles, and the scalar ring is the integers.

Let $G = \mathbb{Z}^2/\Lambda$. Choose representatives (i, j) with $0 \le i < a$, $0 \le j < c$ in lexicographic order. To reduce any integer pair, write $j = qc + r$, $0 \le r < c$, and replace it by $((i - qb) \bmod a, r)$. Let S_1, S_2 be the permutation matrices $(S_k x)_g = x_{g+e_k}$. They commute and are orthogonal. Define

$$A_\Lambda = I + S_1 + S_2, \qquad \mathrm{Fix}_\Lambda = \ker(A_\Lambda : \mathbb{T}^N \longrightarrow \mathbb{T}^N). \tag{4}$$

If two neighbors coincide, their contributions add. In particular, at index one the matrix is [3], not [1]. Equation (4) is exact because a Λ-periodic array is determined by its quotient coordinates and the defining relation must hold at every quotient site.

Theorem 1 (Complete effective fixed-group presentation)**.** *Suppose integer unimodular matrices U, V put A_Λ into Smith form*

$$UA_\Lambda V = \mathrm{diag}(d_1, \dots, d_r, 0, \dots, 0), \qquad 0 < d_1 \mid \cdots \mid d_r.$$

Then

$$\mathrm{Fix}_\Lambda \simeq \mathbb{T}^{N-r} \times \prod_{j=1}^{r} \mathbb{Z}/d_j\mathbb{Z}. \tag{5}$$

This classifies every finite-index lattice fixed group by an explicit integer algorithm. The splitting need not be canonical.

Proof. The matrices U, V induce torus automorphisms since their inverse matrices are integral. A diagonal map $t \mapsto d_j t$ on a circle has kernel $\mathbb{Z}/d_j\mathbb{Z}$, whereas a zero diagonal entry leaves a whole circle free. Applying these facts coordinatewise proves (5). Smith normal form terminates by exact integer Euclidean row and column operations; no numerical eigenvalue threshold enters the statement. □

3 All Hermite lattices and the two resonant modes

Write $\omega = \exp(2\pi i/3)$ and $K = \{(i, j) \in \mathbb{Z}^2 : i - j \equiv 0 \pmod 3\}$. Finite characters of G diagonalize (4); their eigenvalues are $1 + \chi(e_1) + \chi(e_2)$, including every character multiplicity. For unit complex numbers z, w, $1 + z + w = 0$ implies $1 = |1 + z|^2 = 2 + 2\,\mathrm{Re}\, z$. Hence

$$(z, w) \in \{(\omega, \omega^2), (\omega^2, \omega)\}. \tag{6}$$

The first character descends through G exactly when $\Lambda \subset K$; the conjugate has the identical condition.

Theorem 2 (The complete resonance test)**.** *For* (2)*, resonance is equivalent to*

$$3 \mid a, \qquad b \equiv c \pmod 3. \tag{7}$$

At resonance the nullity of A_Λ is exactly two and Fix_Λ *is a finite union of two-tori. Otherwise it is finite, with cardinality*

$$|\,\mathrm{Fix}_\Lambda\,| = |\det A_\Lambda| = \prod_{\chi \in \widehat{G}} |1 + \chi(e_1) + \chi(e_2)|. \tag{8}$$

Proof. The character $(i, j) \mapsto \omega^{i-j}$ is trivial on the two lattice generators precisely under (7). Equation (6) excludes all other zero modes. Fourier diagonalization gives the nullity; Theorem 1 then proves the compact-group assertion. In the invertible case, the number of kernel points is the determinant's absolute value. These arguments reproduce the classical test in [3, Example 3.2] with the current forward-shift convention. □

Index does not decide resonance. For $(a, c) = (3, 1)$, the shear $b = 1$ is resonant and $b = 0, 2$ are not. Thus a census based only on the index loses the continuous fixed stratum. For rectangular lattices $b = 0$, the criterion becomes $3 \mid a$ and $3 \mid c$. Prime indices, composite indices, and neighboring shears use the same source rule; no rational-prime labels are attached to the orbits.

4 Integer covolumes determine the component count

The nonzero Fourier product is finite at resonance, but it is not by itself the order of the torsion cokernel. The following lattice lemma keeps track of the image of all integer vectors.

Lemma 1 (Integer-image covolume formula)**.** *Let $A \in M_N(\mathbb{Z})$ have rank r, and let $P(A)$ be the product of its nonzero singular values. Set $L = \ker A \cap \mathbb{Z}^N$ and $L' = \ker A^t \cap \mathbb{Z}^N$. With Euclidean covolumes in their real spans,*

$$|\operatorname{tors}(\mathbb{Z}^N/A\mathbb{Z}^N)| = \frac{P(A)}{\operatorname{covol}(L)\operatorname{covol}(L')}. \tag{9}$$

Empty products and zero-dimensional covolumes are one.

Proof. The integer kernel L is saturated: if $kz \in L$ for a nonzero integer k, then $Az = 0$. Extend a basis of L to a unimodular integer basis. Put $E = \operatorname{span} L$ and let π be orthogonal projection onto $E^\perp$. The projected complementary columns are a lattice basis of $\pi\mathbb{Z}^N$. The block-volume formula for the extended basis gives

$$1 = \operatorname{covol}(L)\ \operatorname{covol}(\pi\mathbb{Z}^N).$$

The dual lattice of $\pi\mathbb{Z}^N$ in $E^\perp$ is $E^\perp \cap \mathbb{Z}^N$: integrality of inner products against every projected integer vector is equivalent to integrality against every integer vector. This also proves that orthogonal complementary saturated integer lattices have equal covolumes.

The restriction of A from $E^\perp$ to $\operatorname{im} A$ scales r-dimensional volume by $P(A)$. Therefore $A\mathbb{Z}^N = A\pi\mathbb{Z}^N$ has covolume $P(A)/\operatorname{covol}(L)$ in its image span. Its saturation $\operatorname{im} A \cap \mathbb{Z}^N$ has covolume $\operatorname{covol}(L')$ by the complement identity. The index of the integer image in its saturation is precisely the torsion order of the cokernel. Dividing the two covolumes proves (9). The argument does not assume that A is symmetric. □

At resonance, the common real kernel of A_Λ and A_Λ^t consists of vectors with values $A, B, -A-B$ on the three residue classes $i - j = 0, 1, 2 \pmod 3$. The two conjugate Fourier modes prove completeness of this plane. Each class has $N/3$ coordinates. The saturated integer kernel has colour-value basis $(1, 0, -1), (0, 1, -1)$, with Gram matrix

$$\mathcal{G} = \frac{N}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det \mathcal{G} = \frac{N^2}{3}. \tag{10}$$

Both bases are full integer kernel bases, not smaller-index sublattices. Because the shifts commute and are orthogonal, A_Λ is normal; its singular values are the absolute values of its Fourier eigenvalues.

Theorem 3 (Exact resonant components)**.** *Under* (7)*, write* $\operatorname{Fix}_\Lambda \simeq \mathbb{T}^2 \times F_\Lambda$ *with finite* F_Λ*. Its invariant factors are those of Theorem 1, and*

$$|F_\Lambda| = \frac{3}{N^2} \prod_{\substack{\chi \in \widehat{G} \\ 1+\chi(e_1)+\chi(e_2) \neq 0}} |1 + \chi(e_1) + \chi(e_2)|. \tag{11}$$

Proof. Use Lemma 1, the common kernel Gram determinant in (10), and the Smith description of the torus kernel. □

(a,b,c)	N	nullity	nonzero product	components
$(3,1,1)$	3	2	3	1
$(3,0,3)$	9	2	81	3
$(3,0,6)$	18	2	2268	21
$(6,0,6)$	36	2	1778112	4116

The table compares two genuinely different quantities. Treating the fourth column as the fifth fails even at the smallest resonant quotient.

5 A version-specific finite-lattice correction

The accessed arXiv version 0912.5169v1 of [3] prints, in Lemma 2.1 on visibly numbered page 3, a component formula consisting of the uncorrected nonzero Fourier product. The original PDF pages 3 and 4 were rendered and actually inspected; the diagnosis does not rely only on HTML extraction.

Let $\Lambda = K = \langle (3,0), (1,1) \rangle$. On the three quotient sites, the shift matrices are inverse three-cycles and

$$A_K = I + S_1 + S_1^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}. \tag{12}$$

Its integer image is $\mathbb{Z}(1,1,1)$, its cokernel is torsion-free of rank two, and its torus kernel is the connected group $\{(A,B,C) \in \mathbb{T}^3 : A+B+C=0\}$. Its nonzero eigenvalue is 3. The displayed uncorrected formula gives three components; the actual component count is one.

The proof on printed page 4 replaces the image of the whole integer lattice by the image of its intersection with the nonzero-mode real space. For (12), these images are respectively $\mathbb{Z}(1,1,1)$ and $3\mathbb{Z}(1,1,1)$. They are not equal. Lemma 1 uses the projected integer lattice and thereby retains the missing index.

This is a counterexample to that finite-lattice identity in that accessed version. The status of subsequent corrigenda is not established. It is not a refutation of the entropy theorem: for the current source, the logarithmic correction divided by N is $\log(N^2/3)/N$, which tends to zero. The source's Example 3.2 remains the direct owner of the correct resonance condition, and component-growth constructions remain distinct from the ordinary point-cardinality object.

6 Evidence, literature boundary and conclusion

The canonical producer retains every quotient matrix and its integer Smith witnesses. An independent checker imports neither that producer nor a symbolic algebra package: direct multiplication checks the factorization, Bareiss elimination verifies unimodularity, and Faddeev–LeVerrier verifies all characteristic coefficients. A separate library Smith calculation and character-by-character Fourier product give additional controls. The finite grid contains 142 HNF lattices: twenty resonant and 122 nonresonant. It includes every index at most twelve and all shears for $(a,c) = (3,6),(6,3),(6,6)$. These finite receipts test the implementation; the universal theorem is proved independently. Repaired-hash attacks specifically distinguish a pseudodeterminant from a component count, and raw/type-locked evaluation checks reject unknown fields and boolean-to-integer drift.

The direct literature owners must remain visible. Principal-action entropy is an instance of [1]; the exact source resonance is [3, Example 3.2]; the special value belongs to [2]. General periodic-component growth results in [3, Theorem 1.2] are not replaced by a claim about ordinary point-cardinality zeta. The present reconstruction is owner-heavy and is not a novelty certificate. Its version-specific correction is confined to the displayed finite-lattice formula and the explicit integer-image error exhibited above.

The strict Route-A assessment is weak arithmetic relation, weak primitive layer, and failure of the target determinant and target analytic layers. Native commuting Haar Koopman unitaries give only a formal quantum hint; no privileged scalar generator is constructed. No target arithmetic local data, Euler factors, root number, automorphy, target-zero match or Hilbert–Polya operator is claimed. Route B remains disabled. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.

References

[1] D. Lind, K. Schmidt and T. Ward, Mahler measure and entropy for commuting automorphisms of compact groups, *Inventiones Mathematicae* **101** (1990), 593–629. Primary institutional PDF.

[2] C. J. Smyth, On measures of polynomials in several variables, *Bulletin of the Australian Mathematical Society* **23** (1981), 49–63. doi:10.1017/S0004972700006894.

[3] D. Lind, K. Schmidt and E. Verbitskiy, Entropy and growth rate of periodic points of algebraic $\mathbb{Z}^d$-actions, *Contemporary Mathematics* **532** (2010), 195–211. Version-specific formula audit uses arXiv:0912.5169v1, Lemma 2.1, visibly numbered PDF pages 3–4 and Example 3.2.

Round one: integer covolumes and the finite-index correction.